POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2018
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the basic elements of a digital signal processing system? Explain each elements briefly 7
   b) Determine the response of the relaxed system characterized by the impulse response $h[n] = 0.5^n u[n]$ and input $x[n] = 3^n u[n-1]$. 8
2. a) Explain about the Time shifting and Time reversal properties of Discrete Time Fourier Transform. 7
   b) Write down the properties of ROC of Z-transform with examples. 8
3. a) Find the DFT of the sequence 7
   $x(n) = \cos\frac{\pi}{2} n \; ; for \; 0 \leq n \leq 7$
   Using radix–2 Decimation in Time Fast Fourier Transform (DITFFT) algorithm and keep track of all the intermediate quantities by putting them on the diagrams.
   b) Prove that multiplication of two sequences gives the result in circular of these two sequences. 8

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4. a) Draw the lattice-ladder structure for the following IIR system. 7

$$H(z) = \frac{1 - 0.8z^{-1} - 0.15z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

   Is the system stable?
   b) Obtain the direct form I and direct form II realizations for the system described by the following equation: 8
   $y(n) - 3y(n-3) = x(n) + 4x(n-3)$
5. a) Define Digital Filters. Differentiate between IIR & FIR Digital Filters. 8
   b) Design a linear FIR filter using Kaiser window to meet the following specifications: 7

$$0.98 \leq |H(e^{j\omega})| \leq 1.02, \quad for \; 0 \leq |\omega| \leq 0.18\pi$$
$$|H(e^{j\omega})| \leq 0.02, \quad for \; 0.22\pi \leq |\omega| \leq \pi$$

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6. a) Design an FIR filter with 5 coefficients for the following specifications:
   - Passband edge frequency = 0.25 KHz
   - Sampling frequency = 1 KHz
   - Stopband attenuation = 40 dB

   b) Convert the analog filter with system function

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 16}$$

   Into a digital IIR filter by means of the bilinear transformation. The digital filter is to have a resonant frequency of $\omega_r = \pi / 2$.

7. Write short notes on: **(Any two)**
   a) Impulse invariance method for filter design
   b) Frequency response of LTI system.
   c) Energy and Power Signal

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